Animal Rescue Platform

Software Design Documentation

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Contents

1 Introduction

This document describes the design and architecture of the **Animal Rescue Platform**, a web-based system that supports animal rescue organizations in publishing animals for adoption and managing adoption requests.

The documentation follows a structured, engineering-oriented approach and is intended to be understandable for developers, reviewers and stakeholders.

2 Vision and Scope

2.1 Vision

The goal of the project is to provide a reliable, user-friendly and scalable platform that helps animals find permanent homes as efficiently as possible.

2.2 In Scope

- Publishing animals for adoption
- Browsing and filtering animals
- Submitting adoption requests
- Administrative management of content

2.3 Out of Scope

- Payment processing
- Veterinary management
- Multi-organization federation

3 Requirements

3.1 Functional Requirements

- Users can view a list of animals available for adoption.
- Users can view detailed information about an animal.
- Users can submit an adoption request.
- Administrators can create, update and remove animals.

3.2 Non-Functional Requirements

- The system must be SEO-friendly.
- The system must support mobile devices.
- Administrative access must be secured.
- Personal data must be handled in compliance with GDPR.

4 Use Cases

4.1 Use Case Diagram

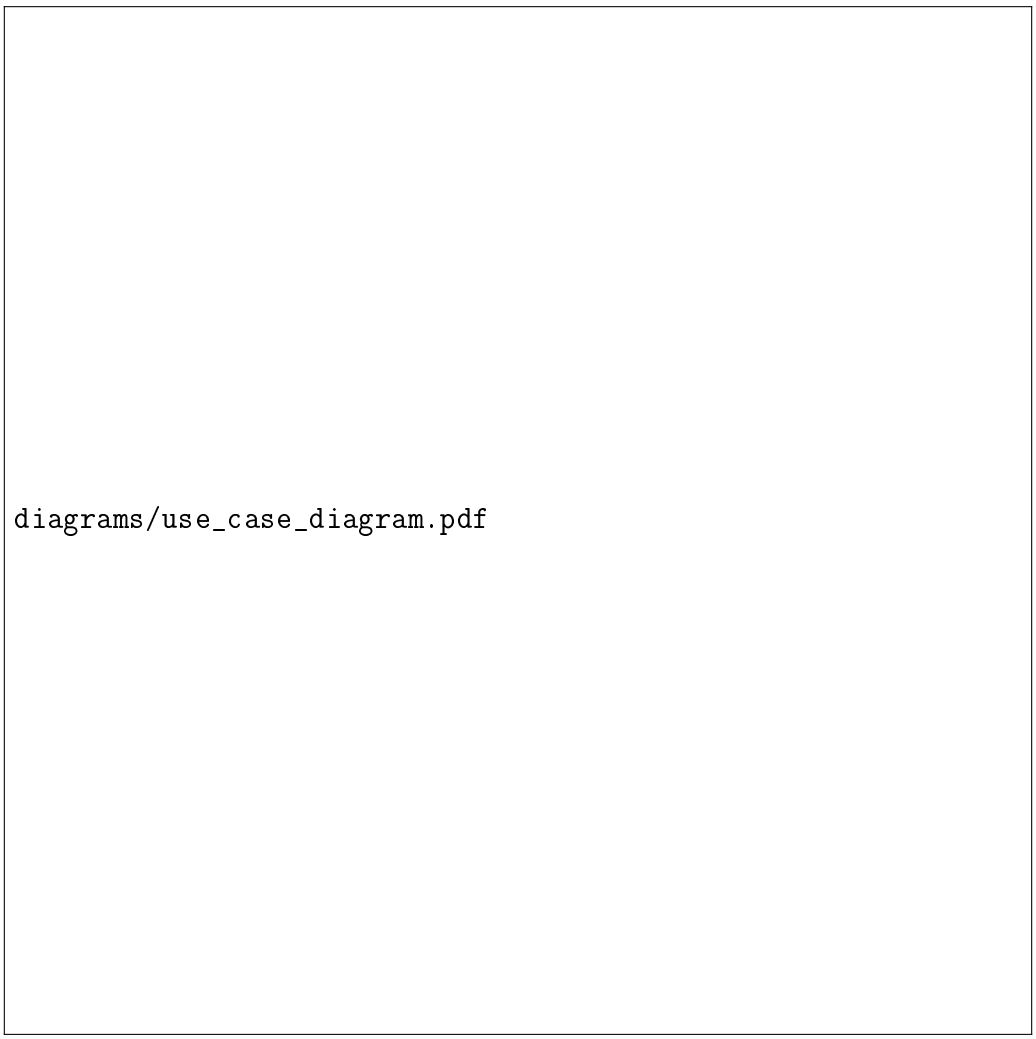


Figure 1: Use Case Diagram

4.2 Use Case Specification: Submit Adoption Request

- **Actor:** Guest User
- **Goal:** Submit a request to adopt an animal
- **Precondition:** Animal exists and is available
- **Main Flow:**
  1. User opens animal detail page
  2. User fills in adoption form
  3. System validates input
  4. Request is stored
- **Postcondition:** Adoption request is created

5 System Architecture

5.1 System Context

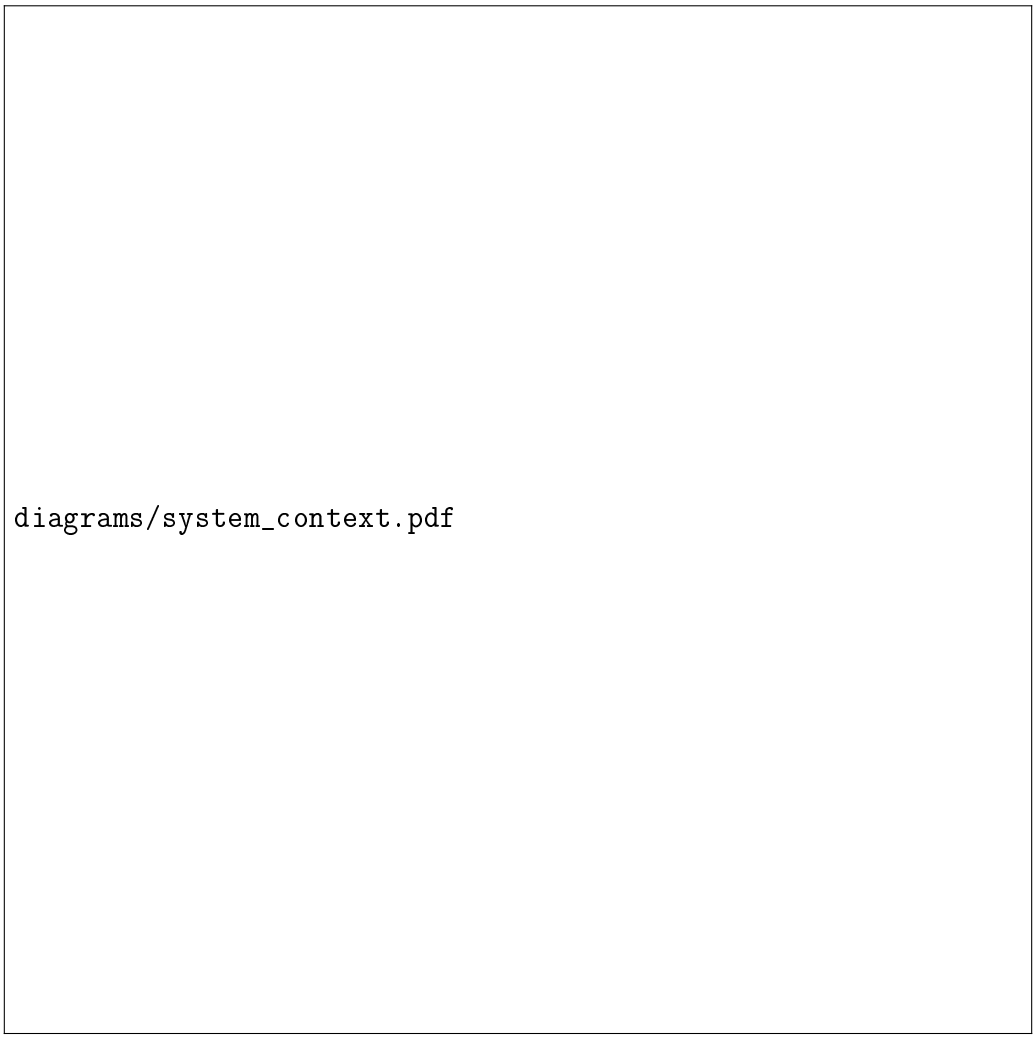


Figure 2: System Context Diagram

5.2 Container Architecture



Figure 3: C4 Container Diagram

6 Backend Design

The backend follows a layered architecture consisting of controllers, services, repositories and security components.

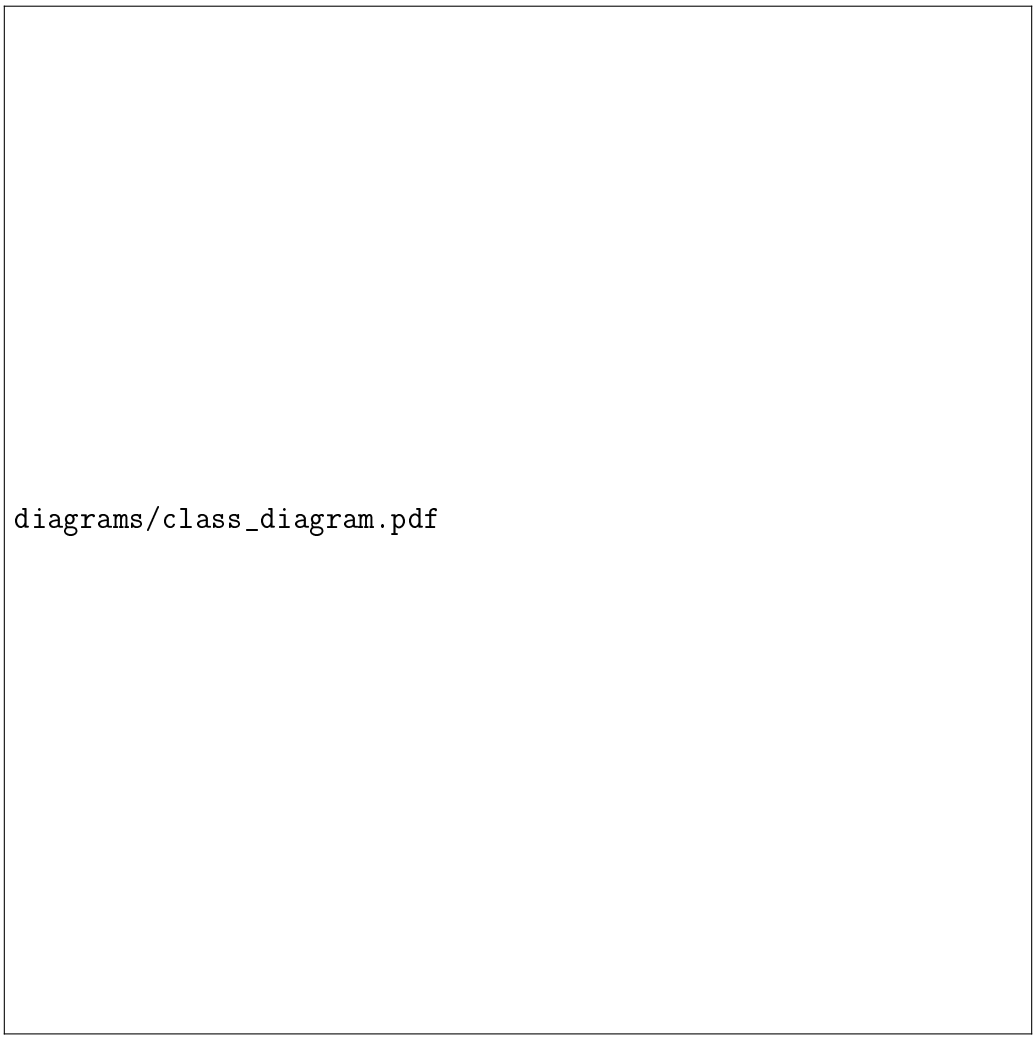


Figure 4: Domain Class Diagram

7 Data Model



Figure 5: Entity-Relationship Diagram

8 API Contract

The system exposes a RESTful API. The full API specification is defined using OpenAPI.
Key endpoints:

- GET /api/animals
- GET /api/animals/id
- POST /api/adoptions

9 Security and Privacy

9.1 Security

The system uses role-based access control and JWT-based authentication for administrative endpoints.

9.2 Privacy and GDPR

Personal data submitted via adoption requests is stored securely and processed according to GDPR requirements.

10 Architecture Decision Records

10.1 ADR-001: Choice of Backend Technology

Spring Boot was chosen due to its maturity, scalability and wide industry adoption.

11 Traceability Matrix

Requirement	Use Case	Implementation
Browse animals	UC-01	GET /api/animals
Submit request	UC-02	POST /api/adoptions

12 Glossary

Animal	A cat or dog published for adoption.
Adoption Request	A request submitted by a user to adopt an animal.
Volunteer	Administrator managing animals and requests.